Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-08-02.jpg":

$$L = \lim_{x \to 0^+} \frac{1}{x} \int_0^x (1 + \sin(2t))^{\frac{1}{t}} \, dt$$

Detailed Solution

To evaluate the limit L, we can recognize this as a classic application of the Fundamental Theorem of Calculus combined with L'Hôpital's Rule, or as the definition of the derivative of an integral function.

Step 1: Apply L'Hôpital's Rule

First, let's rewrite the limit in the form of a fraction:

$$L = \lim_{x \to 0^+} \frac{\int_0^x (1 + \sin(2t))^{\frac{1}{t}} \, dt}{x}$$

As $x \to 0^+$, the numerator $\int_0^x (1 + \sin(2t))^{\frac{1}{t}} \, dt \to 0$ and the denominator $x \to 0$. This gives us an indeterminate form of type $\frac{0}{0}$. We can apply L'Hôpital's Rule by differentiating the numerator and the denominator with respect to x.

By the Fundamental Theorem of Calculus, the derivative of the numerator is the integrand evaluated at x:

$$\frac{d}{dx} \left(\int_0^x (1 + \sin(2t))^{\frac{1}{t}} \, dt \right) = (1 + \sin(2x))^{\frac{1}{x}}$$

The derivative of the denominator is simply 1. Therefore, applying L'Hôpital's Rule gives:

$$L = \lim_{x \to 0^+} \frac{(1 + \sin(2x))^{\frac{1}{x}}}{1} = \lim_{x \to 0^+} (1 + \sin(2x))^{\frac{1}{x}}$$

Step 2: Evaluate the Exponential Limit

We are now evaluating a limit of the form 1^∞. Let the limit be y:

$$y = \lim_{x \to 0^+} (1 + \sin(2x))^{\frac{1}{x}}$$

Take the natural logarithm of both sides:

$$\ln(y) = \ln \left(\lim_{x \to 0^+} (1 + \sin(2x))^{\frac{1}{x}} \right) = \lim_{x \to 0^+} \ln \left((1 + \sin(2x))^{\frac{1}{x}} \right)$$

Using logarithm properties, bring down the exponent:

$$\ln(y) = \lim_{x \to 0^+} \frac{\ln(1 + \sin(2x))}{x}$$

This is another $\frac{0}{0}$ indeterminate form. We can apply L'Hôpital's Rule again. Differentiating the numerator and the denominator:

$$\frac{d}{dx} \ln(1 + \sin(2x)) = \frac{1}{1 + \sin(2x)} \cdot 2\cos(2x) = \frac{2\cos(2x)}{1 + \sin(2x)}$$

$$\frac{d}{dx}(x) = 1$$

Substituting these derivatives back into the limit evaluation:

$$\ln(y) = \lim_{x \to 0^+} \frac{\frac{2\cos(2x)}{1+\sin(2x)}}{1} = \lim_{x \to 0^+} \frac{2\cos(2x)}{1 + \sin(2x)}$$

Now, directly substitute $x = 0$:

$$\ln(y) = \frac{2\cos(0)}{1 + \sin(0)} = \frac{2(1)}{1 + 0} = 2$$

Step 3: Final Calculation

Since $\ln(y) = 2$, we exponentiate to solve for y:

$$y = e^2$$

Thus, our original limit L is equal to e^2.

Final Answer:

$$\mathbf{e^2}$$